

MATPREDICT · MATING-TYPE REPORT

basidio_sporisorium_ab

taxid 999809

Run by MATPredict 0.6.1 · database 2b90ce832408 · NCBI E-utilities · rendered 2026-10-08 04:24 UTC by MATPredict 0.6.1

Result

MATING-TYPE LOCI

undetermined

bLocus

high confidence

undetermined

aLocus

high confidence

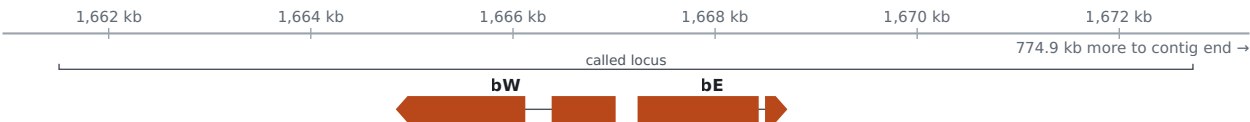
- **undetermined** (bLocus family): MAT gene without flanking genes, high confidence, 11.2 kb on FQ311430.1. Gene order: **bW** · **bE**.
- **undetermined** (aLocus family): complete locus, high confidence, 9.5 kb on FQ311442.1. Gene order: **rba1** · **mfa1** · **pra1**.

Called loci 2

undetermined

bLocus family · Basidiomycota

high confidence



Gene order on FQ311430.1.

core MAT gene

Check: Idiomorph undetermined: the evidence for two idiomorphs is tied or too close to call.

The same genes also match: Basidiomycota:aLocus.

LOCATION	LOCUS	GENES MODELLED
FQ311430.1:1,661,509–1,672,728 · 11.2 kb	A MAT gene with no flanking gene beside it; not a full locus.	2 of 2 gene hits
NEAREST CONTIG END	REGION (SAMTOOLS)	
775.5 kb	FQ311430.1:1661509-1672728	

GENE EVIDENCE							
Gene	Role	Model	Identity %	E-value	Position (strand)	Exons	Reference record
 bW	core MAT	gene model exonerate	80.2	< 1e-180 623 bits	1,664,852–1,667,008 -	2	72559_srs1-h2-8_ bLocus_b1
 bE	core MAT	gene model exonerate	82.0	< 1e-180 731 bits	1,667,242–1,668,705 +	2	72559_srs1-h2-8_ bLocus_b1

undetermined

aLocus family · Basidiomycota

high confidence

760 kb

762 kb

764 kb

766 kb

768 kb

770 kb

called locus

345.4 kb more to contig end →

 **rba1**

 **mfa1**

 **pra1**

Gene order on FQ311442.1.

 core MAT gene

 variable flank

 gene model

 tools disagree

 alternate model

Check:

Idiomorph undetermined: the evidence for two idiomorphs is tied or too close to call.

LOCATION

FQ311442.1:760,387–769,852 · 9.5 kb

LOCUS

Complete locus: MAT gene(s) with their flanking genes.

GENES MODELLED

3 of 3 gene hits

NEAREST CONTIG END

345.9 kb

REGION (SAMTOOLS)

FQ311442.1:760387-769852

GENE EVIDENCE							
Gene	Role	Model	Identity %	E-value	Position (strand)	Exons	Reference record
 rba1	variable flank	gene model exonerate	85.8	1.8e-52 176 bits	760,387–760,794 +	2	5270_521_aLocus_a1
 mfa1	core MAT	gene model exonerate	79.0	9.4e-12 56 bits	764,027–764,140 -	1	49012_cbs-131463_ aLocus_a1
 pra1	core MAT	models differ exonerate miniprot: 767,380–768,583, 80.6%	82.1	1.5e-159 316 bits	767,380–768,571 +	3	49012_cbs-131463_ aLocus_a1

What was searched

ROUTING

By taxonomy.

The genome's taxonomy falls inside a curated family's scope; only those families were searched.

FAMILIES SEARCHED

Basidiomycota:aLocus,
Basidiomycota:bLocus

GENETIC CODE

1

TAXONOMY

NCBI E-utilities

Withheld candidate loci 215

Places with some MAT-gene evidence that did not pass a quality gate. Most genomes have several. They are listed so a known locus can be checked against them, not because they are likely calls.

Reasons: fewer than 2 genes could be modelled (215).

WITHHELD LOCI (215)

family	location	idiomorph	genes	modelled	reason
aLocus	FQ311430.1:9,817–9,918	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:59,010–59,126	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311430.1:287,867–288,025	undetermined	rba1	0	fewer than 2 genes could be modelled
aLocus	FQ311430.1:349,901–350,041	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:551,064–551,222	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:556,743–557,381	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:647,056–647,226	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:801,947–802,036	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:931,687–931,962	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:1,162,291–1,166,348	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311430.1:1,348,077–1,348,187	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:1,355,318–1,355,416	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:1,398,938–1,399,132	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:1,416,362–1,416,703	undetermined	bE, bW	0	fewer than 2 genes could be modelled
aLocus	FQ311430.1:1,507,591–1,507,905	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:1,515,660–1,515,833	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:1,525,350–1,525,484	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:1,591,638–1,591,784	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311430.1:1,661,509–1,672,728	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1:1,684,144–1,684,530	undetermined	bE	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	FQ311430.1: 1,772,727–1,772,876	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1: 1,785,553–1,785,654	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311430.1: 1,811,049–1,811,186	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1: 2,058,651–2,058,761	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1: 2,106,768–2,107,010	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1: 2,196,114–2,196,494	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311430.1: 2,319,042–2,319,230	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:118,672–118,785	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:149,646–149,804	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:157,987–158,178	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:235,998–236,153	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:551,175–551,306	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:582,378–582,563	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:589,465–589,702	undetermined	bE, bW	0	fewer than 2 genes could be modelled
aLocus	FQ311441.1:654,398–654,586	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:716,055–716,147	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:729,480–729,632	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311441.1:741,829–741,954	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:788,144–788,251	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:874,924–875,028	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:895,332–895,493	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:917,817–918,203	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:927,250–927,492	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311441.1:988,597–997,441	undetermined	pra1	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	FQ311441.1:988,597–997,441	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311441.1:1,127,020–1,127,124	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:1,426,386–1,426,532	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:1,447,646–1,450,492	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:1,476,831–1,476,935	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:1,569,123–1,569,338	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:1,583,025–1,583,171	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311441.1:1,617,944–1,618,198	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311433.1:7,285–9,134	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311433.1:15,051–15,176	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311433.1:74,256–74,501	undetermined	pra1	0	fewer than 2 genes could be modelled
aLocus	FQ311433.1:159,299–159,394	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311433.1:176,294–176,536	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311433.1:336,799–336,924	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311433.1:385,881–386,099	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311433.1:445,133–445,327	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311433.1:492,292–492,510	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311433.1:552,195–552,389	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:345,840–345,959	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:397,813–398,007	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:405,777–405,983	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311452.1:427,265–427,441	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:521,491–521,685	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:599,622–599,918	undetermined	bE	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	FQ311452.1:624,393–624,518	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:644,512–644,631	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:725,176–725,454	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:879,943–880,089	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:913,930–914,064	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1:977,052–977,174	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1: 1,023,179–1,026,633	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1: 1,118,332–1,118,442	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1: 1,208,256–1,211,823	undetermined	bE, bW	0	fewer than 2 genes could be modelled
aLocus	FQ311452.1: 1,299,177–1,299,338	undetermined	pra1	0	fewer than 2 genes could be modelled
aLocus	FQ311452.1: 1,317,982–1,318,191	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1: 1,401,356–1,407,425	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	FQ311452.1: 1,451,159–1,451,293	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311442.1:1,355–1,636	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311442.1:17,659–17,784	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311442.1:94,027–94,380	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311442.1:184,665–187,972	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311442.1:194,771–194,896	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311442.1:397,579–397,719	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311442.1:404,790–404,933	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	FQ311442.1:445,248–445,442	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311442.1:503,110–503,235	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311442.1:508,309–509,070	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311442.1:508,309–509,070	undetermined	bW	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes modelled	reason
bLocus	FQ311442.1:698,853–699,071	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311442.1:815,201–815,452	undetermined	bE	0 fewer than 2 genes could be modelled
bLocus	FQ311442.1:855,553–855,903	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311442.1:929,904–930,068	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311442.1:952,184–952,282	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311442.1:959,755–962,347	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311442.1:1,088,117–1,088,242	undetermined	bE	0 fewer than 2 genes could be modelled
bLocus	FQ311438.1:77,661–82,654	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311438.1:120,462–127,323	undetermined	bE, bW	0 fewer than 2 genes could be modelled
bLocus	FQ311438.1:244,453–244,617	undetermined	bE	0 fewer than 2 genes could be modelled
bLocus	FQ311438.1:272,638–272,814	undetermined	bE	0 fewer than 2 genes could be modelled
aLocus	FQ311438.1:324,381–324,554	undetermined	pra1	0 fewer than 2 genes could be modelled
bLocus	FQ311438.1:426,004–426,309	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311437.1:24,397–24,642	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311437.1:37,192–37,308	undetermined	bE	0 fewer than 2 genes could be modelled
bLocus	FQ311437.1:95,886–96,137	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311437.1:181,600–181,722	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311437.1:197,147–197,269	undetermined	bE	0 fewer than 2 genes could be modelled
bLocus	FQ311437.1:224,877–225,128	undetermined	bW	0 fewer than 2 genes could be modelled
aLocus	FQ311437.1:323,881–324,012	undetermined	pra1	0 fewer than 2 genes could be modelled
bLocus	FQ311437.1:339,971–340,144	undetermined	bE	0 fewer than 2 genes could be modelled
bLocus	FQ311437.1:377,114–377,410	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311437.1:389,189–389,368	undetermined	bW	0 fewer than 2 genes could be modelled
bLocus	FQ311440.1:169,373–169,549	undetermined	bE	0 fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	FQ311440.1:446,898–447,017	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311473.1:39,206–39,442	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311473.1:242,682–242,912	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311473.1:261,182–261,298	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311473.1:377,093–377,476	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311473.1:628,613–635,131	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	FQ311463.1:69,337–69,483	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311463.1:91,258–91,476	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311463.1:98,414–101,907	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311463.1:257,379–257,717	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311463.1:424,714–424,815	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311463.1:501,836–501,943	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311463.1:534,393–534,575	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311463.1:540,125–540,217	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311463.1:603,782–603,916	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311463.1:720,496–720,654	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311436.1:121,907–122,170	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311436.1:128,670–133,470	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311436.1:234,661–234,768	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311436.1:385,543–385,728	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311436.1:413,452–413,730	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311436.1:532,568–532,726	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311436.1:553,865–554,029	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311444.1:78,594–78,848	undetermined	bE	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	FQ311444.1:192,644–192,784	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311444.1:349,490–349,876	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311470.1:119,196–119,312	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311470.1:138,653–138,820	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311470.1:245,889–246,104	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311470.1:445,755–445,889	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311470.1:511,435–511,734	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311470.1:528,386–528,622	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311470.1:558,057–558,179	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311474.1:120,097–120,204	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311474.1:208,533–208,646	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311474.1:470,579–473,505	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311474.1:519,182–519,283	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311474.1:528,018–528,500	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311474.1:566,159–566,494	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311474.1:628,914–629,006	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311474.1:639,957–642,916	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	FQ311443.1:148,868–149,014	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311443.1:317,925–318,158	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311443.1:326,890–327,201	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311443.1:395,610–395,828	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:15,283–15,423	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:51,769–51,918	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311471.1:81,453–81,578	undetermined	pra1	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
aLocus	FQ311471.1:200,112–200,231	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:207,247–207,345	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:219,719–219,865	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311471.1:323,757–323,894	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:393,563–393,742	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:563,921–564,037	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:621,737–622,042	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311471.1:670,287–670,385	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:679,339–682,711	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:705,534–705,680	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:923,025–923,237	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311471.1:941,366–941,530	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311432.1:805–933	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311432.1:175,038–175,241	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311432.1:247,880–247,978	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311432.1:403,026–403,151	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311432.1:563,357–563,551	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311432.1:597,776–597,868	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311472.1:33,502–35,438	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311472.1:41,167–41,328	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311472.1:50,546–50,782	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311472.1:95,639–95,854	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311472.1:234,178–234,366	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311472.1:380,159–380,491	undetermined	bE	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
aLocus	FQ311472.1:582,599–582,793	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311472.1:589,078–589,323	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311472.1:758,079–758,339	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311472.1:811,341–815,082	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311469.1:33,171–33,473	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311431.1:379,084–379,293	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311431.1:415,529–415,708	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311431.1:500,463–500,603	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311431.1:588,938–589,168	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311431.1:629,516–629,659	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311435.1:55,475–55,615	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311435.1:86,623–86,775	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311435.1:233,530–233,736	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311435.1:445,000–445,098	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311435.1:540,133–540,279	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311439.1:25,456–25,575	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	FQ311439.1:52,894–53,112	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	FQ311439.1:211,073–211,378	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311439.1:300,864–301,031	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311439.1:401,836–401,934	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	FQ311439.1:492,390–492,536	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	FQ311439.1:498,979–499,236	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311445.1:160,693–160,977	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311434.1:173,047–173,358	undetermined	bE	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	FQ311434.1:352,480–352,629	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	FQ311434.1:434,730–434,891	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	FQ311434.1:541,341–541,712	undetermined	bW	0	fewer than 2 genes could be modelled

Provenance

GENOME FILE	ASSEMBLY	GENOME SHA-256
basidio_sporisorium_ab.fna	27 contigs · 18.5 Mb · N50 772.4 kb	8cb2c5749cc68153beaffb210069f72d07b5b9b11e701f47c5512a3f5e261bab
MATPREDICT	DATABASE	TAXONOMY
0.6.1	2b90ce832408b109 · 163 records	NCBI E-utilities
PARAMETERS	RUN STARTED	RUN TIME
min_hits=2, require_core_role=True, max_polished_clusters_per_family=6, polish_strong_identity=50.0	2026-10-08T04:22:53Z	70 s
FILES		
detection_report.yaml, detected_loci.gff3		
Cite MATPredict as given in CITATION.cff at github.com/stajichlab/MATPredict . The reference records named in the gene tables are curated database entries; each record's metadata cites its source publication.		

How to read this report

Idiomorph

The form of the mating-type locus a strain carries (for example MAT1-1 or MAT1-2, Plus or Minus).

Basidiomycota HD and PR alleles are named after the reference they match.

Low confidence

A single isolated gene hit, or no core MAT gene.

High confidence

Every expected core MAT gene was found and modelled, with a conserved flanking gene where the family has one, on one contig.

Model status

Gene model: the reference protein was aligned to the genome to build the gene (exonerate, miniprot).

Models differ: the two tools disagree on the structure; the alternate model is shown. Hit only: a search hit with no model, weaker evidence.

Medium confidence

The core genes were found, but a gene could not be modelled, no conserved flank was found, or the locus is split (one step down).

Identity

Percent amino-acid identity between the gene (or hit) and the closest reference protein.

Reference record

The curated database record whose protein matched best. Each record's metadata cites the publication or deposit it comes from.

LIMITS

- MATPredict reads an assembly; it is not a mating assay. Two idiomorphs in one assembly can come from a homothallic species, a mixed culture or a merged diploid assembly.
- Fragmented assemblies can split a locus or lose its flanking genes; an uncalled genome is weak evidence of absence.
- Lineages with no curated record are searched with the whole phylum's references; family labels there are less reliable.